

PERSONAL EXPENSE TRACKER

PROJECT REPORT

Python Development Internship Mini Project

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Course: B.Tech in Computer Science & Engineering (AI/ML)

Technology: Python

Database: SQLite

Submitted as part of the Python Development Internship

1. Introduction

The Personal Expense Tracker is a beginner-level Python application developed to record, manage, search, and analyze daily expenses. The application provides a simple command-line interface through which users can enter expense details and perform different expense-management operations.

The project uses SQLite for persistent storage and provides practical experience with Python development, database operations, validation, testing, and project documentation.

2. Project Objective

The main objective of the project is to develop a simple Python application that allows users to record, manage, search, and analyze their daily expenses.

The project also provides practical experience with Python functions, lists and dictionaries, database handling, input validation, exception handling, Git, GitHub, and software testing.

3. Problem Statement

Managing daily expenses manually can make it difficult to keep track of spending. The Personal Expense Tracker addresses this problem by providing a simple application in which expense information can be stored and retrieved when required.

Each expense record contains an ID, title, amount, category, and date.

4. Features

- Add a new expense with title, amount, category, and date.
- View all stored expenses.
- Search expenses by category.
- Calculate total expenses.
- Display a monthly expense summary.
- Calculate yearly expenses as an additional feature.
- Delete an expense using its ID.
- Validate user input.
- Handle common input and application errors.
- Store expense records using SQLite.

5. Technologies Used

Technology	Purpose
Python	Application logic and command-line interface
SQLite	Local persistent data storage
SQL	Database queries and calculations
Git	Version control

Technology	Purpose
GitHub	Source-code repository hosting
Visual Studio Code	Development environment

6. Project Structure

```

Personal-Expense-Tracker/
├──
│   ├── main.py
│   ├── database.py
│   ├── expense_manager.py
│   ├── validator.py
│   ├── requirements.txt
│   ├── README.md
│   └── .gitignore
├──
│   ├── data/
│   │   └── expenses.db
│   └──
│       ├── screenshots/

```

7. Module Description

File	Purpose
main.py	Main application flow, menu, user interaction, and option handling.
database.py	SQLite database connection and table creation.
expense_manager.py	Add, view, search, calculate, and delete expense operations.
validator.py	Validation of title, amount, category, and date.
requirements.txt	Project dependency/environment information.
README.md	Project documentation, setup instructions, usage, and screenshots.

8. Database Design

The application uses an SQLite table named expenses. The table stores each expense as a separate record.

Column	Type	Description
id	INTEGER	Unique expense identifier / primary key
title	TEXT	Name or description of the expense
amount	REAL	Expense amount
category	TEXT	Expense category
date	TEXT	Date of the expense

9. Working of the Application

1. The application starts from main.py and creates the database/table if it does not already exist.
2. The user selects an option from the command-line menu.
3. For adding an expense, the application collects the title, amount, category, and date.
4. The validator module checks the input before the data is stored.
5. The expense manager performs the required database operation.

6. Stored records can be viewed, searched, summarized, or deleted.
7. The application continues showing the menu until the user selects Exit.

10. Input Validation and Exception Handling

Basic validation is included to reduce incorrect data entry. The application checks that the expense title and category are not empty, the amount is numeric and greater than zero, and the date follows the YYYY-MM-DD format.

The application also handles invalid menu choices and common input errors so that an incorrect entry does not unnecessarily terminate the program.

11. Testing

Test Case	Input/Action	Expected Result	Status
Add expense	Enter valid details	Expense stored successfully	Pass
View expenses	Select View All	Stored records displayed	Pass
Search	Enter a category	Matching records displayed	Pass
Total	Select Total Expenses	Total amount displayed	Pass
Monthly summary	Enter year/month	Monthly total displayed	Pass
Delete	Enter existing ID	Expense removed	Pass
Invalid amount	Enter text/negative value	Validation message displayed	Pass
Invalid date	Enter incorrect format	Validation message displayed	Pass

12. Project Screenshots

Insert the actual screenshots from the project in this section before submitting the final report. Recommended screenshots are:

- Figure 1: Main menu – 01_main_menu.png
- Figure 2: Adding an expense – 02_add_expense.png
- Figure 3: Viewing all expenses – 03_view_expenses.png
- Figure 4: Searching expenses by category – 04_search_expense.png
- Figure 5: Expense summary – 05_expense_summary.png
- Figure 6: Deleting an expense – 06_delete_expense.png
- Figure 7: Input validation – 07_validation.png

13. Challenges Faced

During development, attention was required for organizing the application into separate modules, connecting the Python code with SQLite, validating user input, and testing database operations. Testing each menu option individually helps identify and fix errors before final submission.

14. Future Improvements

- Add an option to edit an existing expense.
- Export expense records to CSV.
- Add graphical charts for spending analysis.
- Add budget limits and alerts.
- Add a graphical user interface.
- Add authentication if the application is later extended to multiple users.

15. Conclusion

The Personal Expense Tracker provides a simple way to record and analyze daily expenses through a Python command-line interface. The project demonstrates practical use of Python functions, SQL queries, SQLite data storage, input validation, exception handling, testing, and project organization.

The project also provides a foundation for adding more advanced expense-management features in the future.

16. Final Submission Checklist

- ■ GitHub Repository
- ■ Complete Source Code
- ■ README.md
- ■ Project Report
- ■ Project Screenshots
- ■ Final Project Demo

17. Project Links

GitHub Repository: _____

Final Project Demo: _____

18. Author

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Internship: Python Development Internship